

BUSA2001 – Assignment 1

1. Descriptive data

The insurance dataset includes a variety of variables that provide information on some aspects contributing to the variation in insurance charges. Specifically, there are six independent variables, including both categorical variables, such as gender, smoking status, and region, as well as continuous variables, such as age, BMI, and number of children, that aim to predict the outcome of a continuous dependent variable, insurance charges. These factors provide the analysts with comprehensive demographic and health-related data, which may make it ideal for analysing risk factors and demonstrating the accuracy and reliability of the predictive models for insurance charges.

The major goal of this analysis is to determine how each of these factors interact and impact final insurance charges. First, we must distinguish between the dataset's regression and classification tasks. Regression analysis focuses on continuous variables like age, BMI, number of children, and target variable costs which aims to predict the continuous outcome of insurance charges using the values of the independent variables. However, the categorical variables, such as gender, smoking status, and region, require a classification approach. These variables must be encoded into numerical representations before being used in machine learning, allowing us to incorporate these categorical features into the regression analysis to ensure accurate predictions about insurance charges.

Therefore, based on the factors mentioned that explain the reason why this is an appropriate dataset for the analysis. Furthermore, after importing the dataset, I check for missing values and information to ensure the dataset is reliable for analysis by utilising the `isnull()` and `describe()` functions, resulting in a clean dataset for analysis with no missing value or null count.

2. Data visualisation



Figure 1

From figure 1, the distribution shape of the charges is right-skewed, indicating that most of the data points are concentrated on the lower end, closer to 0, with fewer data points at higher charge values. Most charges are

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below 10,000, suggesting lower charges are more common within this dataset. Some higher charge values are present but much less frequent, as evidenced by the long tail extending to the right. These outliers likely represent exceptional cases or rare instances of high charges.

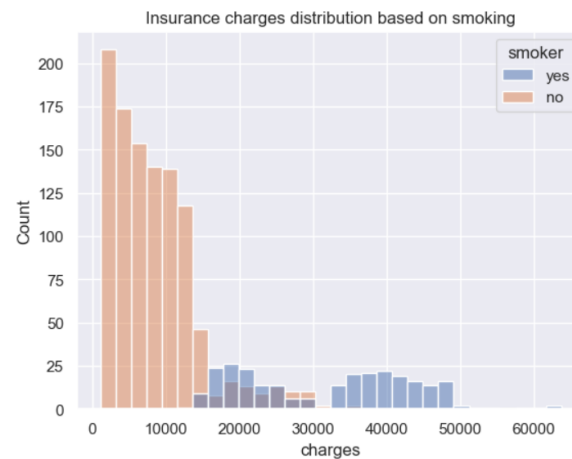


Figure 2

More specifically, as seen in figure 2, the insurance costs in the range of 10,000 to 16,000 mainly belong to non-smoking insurance buyers. The remaining insurance costs and rare instances of insurance charges from 20,000 or more belong to smoking insurance buyers, so it can be commented that smoking affects insurance rates because it has a direct impact on human health.



Figure 3

According to the data in figure 3, it is considered that there is a moderate relationship between the three factors: the higher the age of the individuals, the higher the insurance charges are in both smoking statuses; however, the observation with 'yes' status tend to pay higher insurance charges compared to that with 'no' status in different ages.

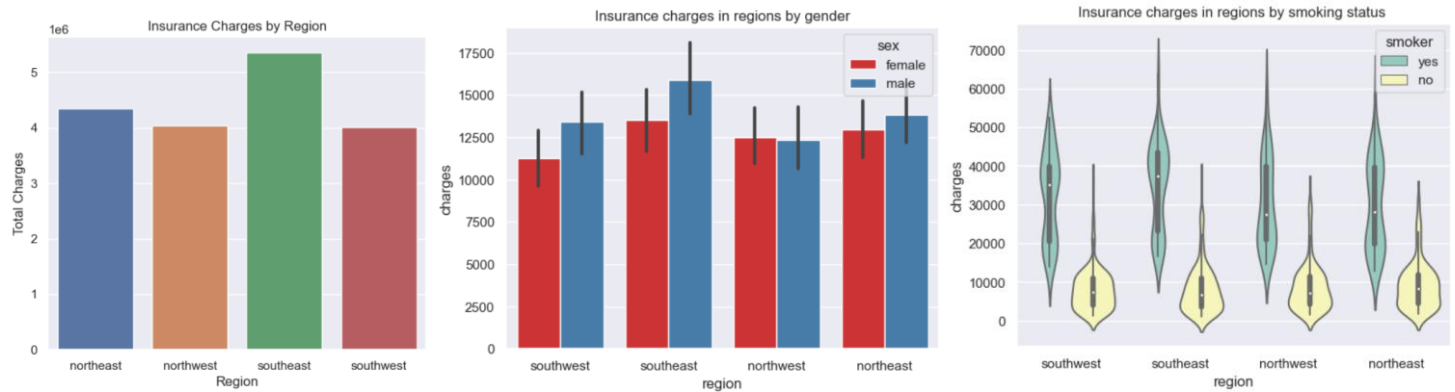


Figure 4

Figure 4 shows that the southwest has the greatest total insurance costs, followed by the northeast, and finally by the northwest and southwest, which have about identical amounts. The violin plot shows that the greatest charges for smoking are in the South-east, while the lowest are in the North-east. People in the South-west tend to smoke more than those in the North-east, although altogether, the North-east has higher charges by gender than the South-west and North-west.

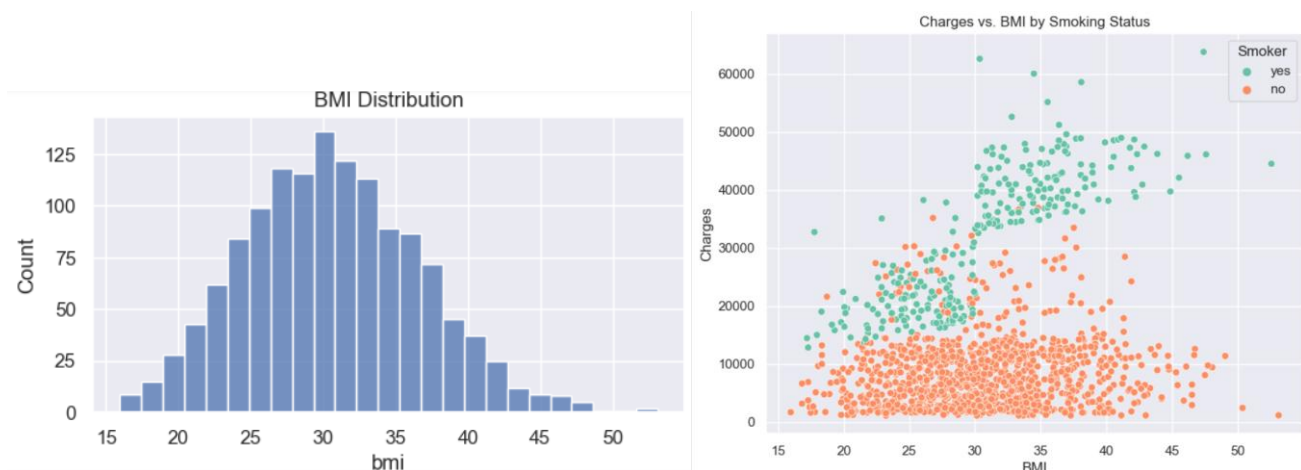


Figure 5

According to the chart of data 5, it is clear that the distribution of BMI in the collected surveys is that 1/5 have a BMI below 25, mainly focusing on fluctuating from about 25 and above, and people around 35 gradually decrease, and the maximum recorded is about 53. There is a positive correlation in smoking status with a strong linear relationship between BMI and charges, meaning that the index of BMI increases, resulting in an increase in insurance charges. While this trend is not clearly shown in non-smoking status.

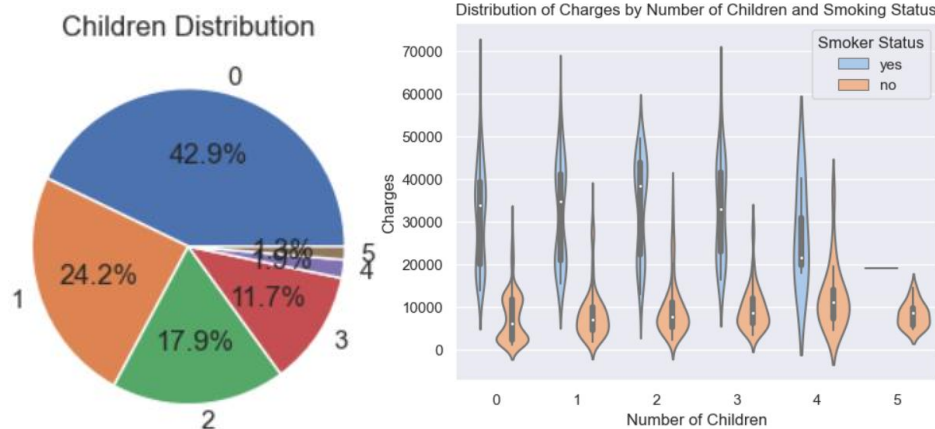


Figure 6

The majority of individuals in the dataset have smaller family sizes. The largest group, at 42.9%, consists of individuals with no children, followed by 24.2% with one child and 17.9% with two children. Only 11.7% have three children, and 1.3% and 1.2% have four and five children, respectively. The family that has more children are generally smokeless than the family with fewer children, as shown in Figure 6 in the violin plot.

3. 'Groupby' functions

```
#SMOKING
# Average the insurance charges by smoker
ave_charges_bysmoker = df_pl.groupby('smoker').agg(pl.col('charges').mean()).alias('Average charges by smoker')
print(ave_charges_bysmoker)
# The different charges distribution based on smoker
tl_charges_bysmoker = df_pl.groupby('smoker').agg(pl.col('charges').sum()).alias('Total charges by smoker')
print(tl_charges_bysmoker)
```

shape: (2, 2)

smoker	Average charges by smoker
---	---
str	f64
yes	32050.231832
no	8434.268298

shape: (2, 2)

smoker	Total charges by smoker
---	---
str	f64
no	8.9741e6
yes	8.7818e6

This notable difference indicates that smoking significantly raises the cost of insurance due to associated health risks compared to non-smoking. However, it is the opposite as there are more non-smokers in the dataset, and the overall total insurance charges are slightly higher for non-smokers.

```
#BMI
# Category the BMI into 3 types and calculate mean of each type
underweight_df = df_pl.select(pl.col(['bmi' , 'charges']).filter(pl.col('bmi')<=18.5))
underweight_df
ave_charges_underweight = underweight_df.select(pl.col('charges').mean().alias('Average charges for underweight'))
print(ave_charges_underweight)

overweight_df = df_pl.select(pl.col(['bmi' , 'charges']).filter(pl.col('bmi')>25))
overweight_df
ave_charges_overweight = overweight_df.select(pl.col('charges').mean().alias('Average charges for overweight'))
print(ave_charges_overweight)

normalweight_df = df_pl.filter((pl.col('bmi') > 18.5) & (pl.col('bmi') <= 25)
                               ).select(['bmi', 'charges'])
ave_charges_normalweight = normalweight_df.select(pl.col('charges').mean().alias('Average charges for normalweight'))
print(ave_charges_normalweight)
```

shape: (1, 1)

Average charges for underweigh...

f64
8657.620652

shape: (1, 1)

Average charges for overweight

f64
13946.476035

shape: (1, 1)

Average charges for normalweig...

f64
10435.440719

Because many BMI indexes are provided in the dataset, I divided it into 3 main categories, including normal-weight, underweight, and overweight. Then I calculated the average charges for each category for comparison purposes, such as which category mostly influences the insurance charges.

```
# AGE
under25_df = df_pl.select(pl.col(['age' , 'charges']).filter(pl.col('age')<=25))
under25_df
ave_charges_under25 = under25_df.select(pl.col('charges').mean().alias('Average charges for under25'))
print(ave_charges_under25)

under35_df = df_pl.select(pl.col(['age' , 'charges']).filter((pl.col('age') > 25) & (pl.col('age') <= 35)))
under35_df
ave_charges_under35 = under35_df.select(pl.col('charges').mean().alias('Average charges for under35'))
print(ave_charges_under35)

under45_df = df_pl.select(pl.col(['age' , 'charges']).filter((pl.col('age') > 35) & (pl.col('age') <= 45)))
under45_df
ave_charges_under45 = under45_df.select(pl.col('charges').mean().alias('Average charges for under45'))
print(ave_charges_under45)

under55_df = df_pl.select(pl.col(['age' , 'charges']).filter((pl.col('age') > 45) & (pl.col('age') <= 55)))
under55_df
ave_charges_under55 = under55_df.select(pl.col('charges').mean().alias('Average charges for under55'))
print(ave_charges_under55)

over55_df = df_pl.select(pl.col(['age' , 'charges']).filter((pl.col('age') > 55) ))
over55_df
ave_charges_over55 = over55_df.select(pl.col('charges').mean().alias('Average charges for over55'))
print(ave_charges_over55)
```

shape: (1, 1)

Average charges for under25

f64
9087.015807

shape: (1, 1)

Average charges for under35

f64
10495.163047

shape: (1, 1)

Average charges for under45

f64
13493.485247

shape: (1, 1)

Average charges for under55

f64
15986.90031

shape: (1, 1)

Average charges for over55

f64
18795.993816

Because there are so many different ages in the data, I re-divided it into the five types mentioned above. I calculated the average insurance premium of each type, from which readers can determine the age level with the amount of insurance charges, such as what age groups have the highest or the lowest insurance charges.

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4. Machine learning (linear regression)

	age	sex	bmi	children	smoker	region
0	19	1	27.900	0	1	1
1	18	0	33.770	1	0	0
2	28	0	33.000	3	0	0
3	33	0	22.705	0	0	3
4	32	0	28.880	0	0	3

charges

0	16884.92400
1	1725.55230
2	4449.46200
3	21984.47061
4	3866.85520

X_train: (936, 6)

X_test: (402, 6)

Y_train: (936, 1)

Y_test: (402, 1)

the intercept value: [-13299.64129718]

the coefficient for each variable: [[253.13748879 227.57372091 349.69180078 579.10466192
24090.324089 199.46704261]]

R^square for training data: 0.7574340926271486

R^square for testing data: 0.7318630535235593

The linear regression model:

$$Y (\text{insurance charges}) = -13299.62127 + 253.13748879(\text{age}) + 227.57372091(\text{sex}) + 349.69180078(\text{bmi}) + 579.10466192(\text{children}) + 24090.324089(\text{smoker}) + 199.46704261(\text{region})$$

From the findings above, all these features affect the final insurance charges, especially the smoker (smoking status), which causes the most significant influence, showing that smokers dramatically raise the insurance charges compared to other factors. The R^square value for both training and testing data indicates a good fit on both sets; however, the slight decrease from the training set to the testing set means that the model generalized well but not perfectly.